

EDUCATION

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- **STEM High School for Boys** 6th of October City, Egypt
 - *STEM High School; GPA: 4.0/4.0 (unweighted)*
 - **SAT:** 1450
 - **School Coursework:** English, German, Mathematics, Engineering Mechanics, Physics, Chemistry, Arabic, Biology, Earth Science, Capstone (engineering projects), Computer Science
 - **Additional Coursework:** October Physics Club: General physics for engineers and scientists; Science Internship Program: Basics of electrical engineering, wiretap basics, wireless communication basics, and an introduction to machine learning; Harvard CS50: Introduction to artificial intelligence, Python, C, and C++; October Math Club (Calculus Track): Calculus I & II; PennX Robotics (Dynamics & Control): robot-arm dynamics/modeling, linear control, state-space & MIMO systems, and nonlinear control; LaunchX: Entrepreneurship (Becoming an Entrepreneur) and Marketing

SKILLS SUMMARY

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- **Programming:** C, C++, Python, HTML, JavaScript, SQL, Java
 - **Machine Learning / Data:** PyTorch, TensorFlow, Pandas, OpenCV
 - **Engineering & Prototyping:** Electronics, hardware prototyping, microcontrollers, embedded development, engineering design, troubleshooting
 - **Academic & Technical:** Research methods, technical/academic writing, computational physics, $\text{\LaTeX}$
 - **Professional Soft Skills:** Leadership, time management, communication, collaboration, mentoring, public speaking
 - **Languages:** English (C1), German (A2)

EXPERIENCE AND PROGRAMS

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- **Science Internship Program (SIP), University of California, Santa Cruz** Santa Cruz, CA
 - *Electrical Engineering Track (Intern)* Summer 2025
 - **Scholarship:** Admitted through a highly competitive selection process and awarded a full scholarship (\$12,000).
 - **Team Leadership:** Led a team developing a deep-learning autoencoder for physical-layer security over wiretap channels.
 - **Implementation & Evaluation:** Implemented and evaluated models in Python/TensorFlow, including performance analysis across varying SNR conditions.
 - **Training:** Completed 75+ structured sessions plus individualized mentoring; strengthened foundations in wireless communication and electrical engineering.
 - **Yale Young Global Scholars (YYGS)** Yale University (In person) 2024
 - *Participant — Politics, Law, and Economics (PLE) Track*
 - **Scholarship:** Awarded a \$6,600 scholarship.
 - **Academic Engagement:** Completed 7 PLE lectures and 8 seminars focused on core political, legal, and economic challenges.
 - **Leadership:** Developed and presented a political platform; won the program election as the highest vote-getter.
 - **AI & Decision-Making:** Analyzed how AI can influence political decision-making and evaluated strategies for addressing key PLE challenges.
 - **Accelerate College Program, Northeastern University** Boston, MA / In person 2024
 - *Participant — Engineering Innovation & Design Track*
 - **Scholarship:** Awarded a \$5,000 scholarship.
 - **Autonomous Vehicle Project:** Directed a team project to design an autonomous car, including CAD/CAM workflows, 3D printing and laser-cut fabrication, and Arduino-based control, following 30 engineering sessions.
 - **LaunchX** Online 2025
 - *Participant — Entrepreneurship Program; Founder & Web Developer (Mindpate)*
 - **Venture Execution:** Led a team bootstrapping Mindpate from idea to revenue launch: 100 sessions, negotiated with business professionals, generated \$250+ revenue; owned product and go-to-market strategy.
 - **Echolex** Egypt / Remote 2024
 - *Founder / Developer*
 - **Product:** Developed an English-learning platform featuring an AI tutor and real-time caption generation; built in collaboration with an OpenAI mentor.

LEADERSHIP AND SERVICE

- **STEM October Physics Club** Egypt
EPhO Preparation Member & Mentor & Problem-Solving authore for IPhR Competition 2024, 2025
 - **Role:** Academic Mentor for 80 students, Led structured problem-solving, and preparation efforts for high-level physics competitions.
- **Yale Young Global Scholars (YYGS)** Global
Ambassador 2025
 - **Outreach:** Authored guides and blog posts, produced videos, and delivered 7 workshops to 200+ students.
 - **Mentorship:** Supported peer applicants; contributed to two successful admissions outcomes.
- **October English Club** Egypt
Academic Writing Mentor 2025 (update)
 - **Teaching:** Led 10 instructional sessions for 100 students in Grade 10–11 on academic writing using structured curricula and Oxford-style academic writing resources.
- **COMPASS (October Physics Club)** Egypt
Research Participant 2024 – 2025
 - **Computational Physics:** Modeled, simulated, and analyzed systems in classical mechanics and statistical mechanics using Python, emphasizing problems difficult to solve analytically.
- **Elshams Club Running Team** Cairo, Egypt
Founder 2023 – Present
 - **Community Building:** Led a 60-member weekly running program; raised \$400/year for transportation and venue costs.
 - **Event Operations:** Organized 4 running events, including races at heritage sites and Cairo Stadium, serving 200 runners.

HONORS AND AWARDS

- Gold Medal, Egypt International Science and Technology Fair (EISTF): 1st place among 1,600 teams (top 0.063%).
- National Finalist, International Science and Engineering Fair (ISEF): top 1% among 1,000 teams.
- Future Innovator Scholarship: \$3,590 (top 0.70% of 1,400) for an engineering summer program at Oxford University.
- National Finalist, Egyptian Physics Olympiad: top 1.03% among 970 students.
- Awarded \$10,000 in funding for the Echoland Gloves project following evaluation at the Science Land Fair (Cairo, 2025).
- Ranked 1st / 130,000 in Grade 9 Cairo national exams (top 0.00077%); selected top 120 / 25,000 in STEM school qualification exams (IQ, Math, English, Science).
- Egypt's sole nominee for the China Adolescents Science and Technology Innovation Contest.
- Ranked in the top 5% at Y Combinator, out of 27,000+ teams.

SELECTED PROJECTS

- **Echoland Gloves (Accessibility Wearable for Deaf/Hard-of-Hearing Users):** Built a glove-based system to improve two-way communication: (1) sign-to-speech via sensing and classification of sign-language hand movements into audible speech, and (2) speech-to-text transcription displayed on an LCD for the deaf user. Emphasized end-to-end accessibility: sensing, inference, and user-facing output for real-world interaction.
- **Arabic Media to English Translation Platform (AI + APIs):** Developed a system to translate complex Arabic content (e.g., poetry and multimedia) into accessible English while preserving meaning and cultural nuance. Implemented context-aware translation plus post-editing to resolve idioms, metaphors, and allusions; outputs include readable prose/subtitles with optional literal glosses.
- **Telegram Chatbot for Arabic Learners:** Built a tutoring chatbot supporting text and voice input: speech transcription, dialogue-state tracking via lightweight memory, and responses in text or synthesized audio. Implemented role-aware prompting and robust routing/handlers (e.g., onboarding, message processing, and interaction flows).

RESEARCH

- **Deep Learning for Physical-Layer Security (Wiretap Channels)** —
Research Focus 2025
 - **Objective:** Investigated secure communication using neural autoencoders that maximize reliable decoding for a legitimate receiver while limiting information available to an eavesdropper.
 - **Methods:** Simulated Gaussian noise channels across varying SNR regimes; evaluated BER, BLER, and leakage upper bounds.
 - **Optimization:** Tuned loss-weight strategies to quantify and control reliability–secrecy trade-offs.
- **Extended N-Body Dynamics (Four-Body Extension of the Three-Body Problem)** —
Research Paper 2024
 - **Contribution:** Authored a paper extending the classical three-body formulation by introducing a fourth body and analyzing perturbations in velocity and energy profiles.